

HW5

a. $I = \frac{1}{2} k_n' \frac{W}{L} (V - V_t)^2$

$$(V - V_t)^2 = \frac{2IL}{k_n' \cdot W}$$

$$V - V_t = \sqrt{\frac{2IL}{k_n' \cdot W}}$$

b. $V_{ov1} = 0.6 - 0.4 = 0.2V$

$$V_{ov2} = 1.3 - 0.6 - 0.4 = 0.3V$$

$$V_{ov3} = 2.1 - 1.3 - 0.4 = 0.4V$$

$$\frac{W}{L} = \frac{2I}{k_n' (V_{ovi})^2}$$

$$\left(\frac{W}{L}\right)_1 = \frac{2 \times 0.2}{0.4 \times (0.2)^2} = 25$$

$$\left(\frac{W}{L}\right)_2 = \frac{2 \times 0.2}{0.4 \times (0.3)^2} = 11.1$$

$$\left(\frac{W}{L}\right)_3 = \frac{2 \times 0.2}{0.4 \times (0.4)^2} = 6.25$$